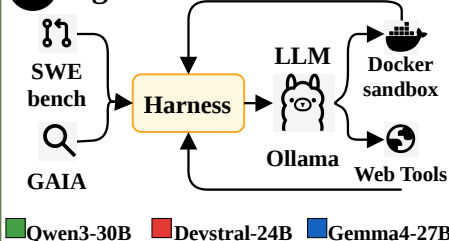
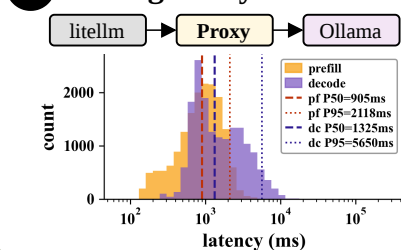


A. Trace Construction

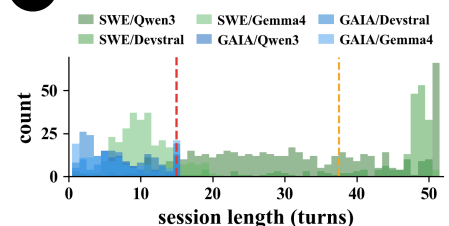
A1 Agent Harness



A2 Timing Proxy



A3 Data Profile



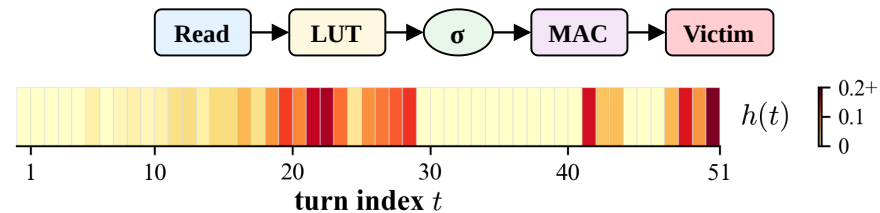
1,451 sessions · 33,596 turns

Dataset	Model	Sessions	Turns
SWE-bench	Qwen3-30B	475	11,280
GAIA	Qwen3-30B	220	5,300
SWE-bench	Devstral-24B	450	10,650
GAIA	Devstral-24B	215	5,160
SWE-bench	Gemma4-27B	490	11,666
GAIA	Gemma4-27B	245	5,840

B. Algorithm Design

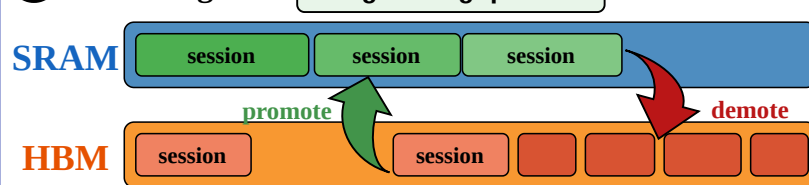
B1 SPEAR Scorer

$$\text{score} = g/\sigma(t) + (1-\sigma) \cdot P$$



B2 TIDE Migrator

$$\text{budget} = \text{gap} \times \text{bw}$$



B3 UNISON Event Pipeline

on request

REQ

replayable traces with policy

SPEAR

events

ALLOC

INFER

EMA

GAP

events

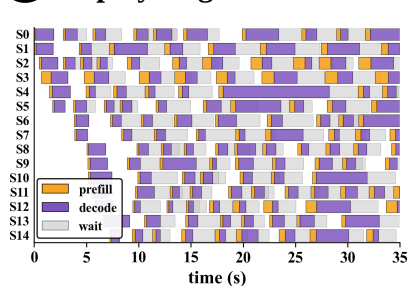
on gap

TIDE

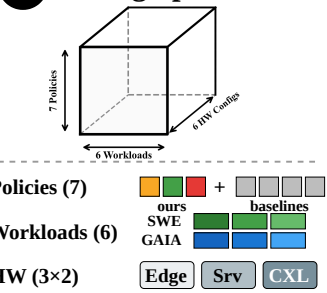
END

C. Evaluation Framework

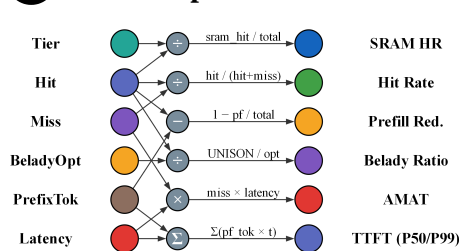
C1 Replay Engine



C2 Config Space



C3 Metric Pipeline



C4 Evaluation Rigor

